Product Deconstruction of Empirio and System Architecture for an Autonomous, Zero-Login Business Case Simulation Engine

Building a free, zero-login business simulation application that improves upon **Empirio: Learn Business** requires pairing Empirio's core pedagogical loop—stepping into high-stakes executive crossroads—with an automated, source-verified data ingestion and scenario-synthesis backend. Released in March 2026 by the Romanian-American edtech studio **Smartheads** (operated by **Deepstash SRL** and **Brainstash, Inc.**), Empirio scaled to over 56,000 Android installs by early November 2026 and an estimated $16,400 to $133,700 in gross monthly iOS App Store revenue in October 2026 (according to AppGazers model estimates). It achieves this through a 6-step interactive case format that evaluates players across 10 psychometric dimensions of financial thinking. However, user reviews across Google Play, the Apple App Store, and Trustpilot expose critical product vulnerabilities in Empirio: mandatory email verification loops that lock out users, subscriptions costing up to $89.99/year, bandwidth-heavy AI videos prone to visual inaccuracies, unverified claims lacking source citations, and narrative pacing that telegraphs quiz answers before decision checkpoints.

To eliminate these failure modes while keeping operational and server costs near zero for unauthenticated users, your application should replace the traditional client-server user database with a **Static-Edge, Local-First Architecture**. In this design, the mobile and web clients never query a live application database or require user accounts. Instead, user state—including decision history, streaks, and the 10-dimension psychometric vector—is persisted directly on the user's device using synchronous memory-mapped key-value storage (react-native-mmkv) for instant UI state hydration alongside embedded relational storage (op-sqlite, expo-sqlite, or IndexedDB) for querying historical case logs. Cross-device continuity can be handled anonymously through compressed, client-encrypted state payloads exported via QR codes or shareable recovery strings.

On the backend, content generation operates strictly as an asynchronous, scheduled batch pipeline decoupled from user traffic. Scheduled workers continuously ingest structured regulatory filings (SEC EDGAR Forms 10-K, 10-Q, and 8-K via edgartools), corporate press wires, and historical business archives. High-speed heuristic content extractors (rs-trafilatura) strip HTML boilerplate before passing raw texts into an entity-and-timeline extraction pipeline. A multi-stage Large Language Model (LLM) synthesis engine—governed by deterministic validation gates that filter out outcome bias and answer leaks—compiles these events into standardized JSON game bundles with source citations. These static JSON bundles and delta manifests are published to **Cloudflare R2** object storage, leveraging its permanent free tier (10 GB storage, 1 million Class A writes, 10 million Class B reads per month, and zero egress fees) to serve client reads globally without runtime compute overhead.

1. Deep Research & Product Teardown: Empirio

Disambiguation and Corporate Lineage

Before examining the product mechanics, it is necessary to distinguish between two unrelated software projects sharing the name "Empirio." A prototype browser-based multiplayer city-builder and empire strategy game (game.empirio-game.pp.ua) surfaced on Reddit's r/CityBuilders community, supporting up to 10 players per world. The target of this architectural study—and the application matching the yellow crown-and-sunglasses mascot (#F5B21F brand hex code paired with #000000 black)—is **Empirio: Learn Business** (com.empirioapp on Google Play; App ID 6759285666 on the Apple App Store).

Empirio is part of a unified micro-learning app portfolio operated by joint data controllers **Deepstash SRL** (headquartered at 47 Vespasian Street, Room 3, District 1, Bucharest, Romania, serving as the operational and technical controller) and its US parent, global distributor, and billing entity, **Brainstash, Inc.** (registered in Newark, Delaware). Founded in 2018 and backed by Eastern European venture capital firm GapMinder Ventures with a Dealroom-reported $50 million valuation, Deepstash rebranded its multi-app suite to **Smartheads** in 2026 after surpassing 10 million cumulative users with an 80+ person team in Bucharest.

Smartheads operates a multi-app ecosystem across five personal development verticals that share an underlying technical architecture, cross-app analytics engine, and GDPR data controller framework:

1. **Deepstash: Smarter Every Day!** — Book summaries and curated micro-ideas in 5 minutes a day (rated 4.6 stars across 205,000+ Google Play reviews).
2. **Paladin: Learn History** — Narrative-driven interactive historical lessons, which became the company's second-largest product in 2026 (4.6 stars on Google Play; 4.2–4.3 on Trustpilot across ~355–367 reviews).
3. **Dilemmo: Make Better Decisions** — Interactive moral, ethical, and real-life decision simulations that generate psychometric profiles (4.2–4.3 stars on Google Play with 100,000+ downloads).
4. **Protocol: Health & Longevity** — Habit tracking and longevity optimization (4.4–4.5 stars on Google Play).
5. **Empirio: Learn Business** — Executive strategy, finance, and historical corporate case simulations.

Platform Availability, Growth Metrics, and Monetization

Launched in March 2026 (releasing on the iOS App Store on March 3, 2026, and on Google Play in March 2026), Empirio iterated through more than 20 point releases to reach version 2.3.0 by early November 2026. The application is available on Android (requiring Android 7.0 / API 24 or higher, compiled for arm64-v8a), iOS, iPadOS, tvOS, and Apple Vision. On Android, the binary size ranges from 82.31 MB on Google Play to 124.5–139.51 MB across APK/XAPK distributions, incorporating 94 software libraries and requesting 41 system permissions (including audio recording, external storage, biometric/fingerprint hardware, billing, and Google AdServices attribution IDs).

User acquisition is heavily driven by paid social advertising. AdScan tracking of the Meta Ad Library recorded 1,368 Facebook ads for empirioapp.com, led by the creative hook *"MBA secrets in 5 minutes a day"*, whose most-seen variant achieved a reach of 289,300 users over a 56-day run. By November 2, 2026, AppBrain tracked 56,000+ total installs on Google Play, with

26,000 installs occurring in the preceding 30 days (~879–880 daily installs), ranking the app #103 in US Education with a 4.57-star average across 2,193 ratings (1,600 five-star, 380 four-star, 190 three-star, 0 two-star, and 39 one-star ratings; Google Play's localized storefront also displays up to 4.8 stars across 2,280 reviews). On the US Apple App Store, the app holds a 4.6-star rating across 649–713 ratings, ranking between #57 and #108 in Education, while Trustpilot records a 4.5 TrustScore (4.6 stars) across 86–92 reviews. According to third-party estimates from AppGazers (using an 80% confidence band for October 1–31, 2026), worldwide iOS App Store downloads ranged from 13,000 to 68,900 for the month, generating an estimated $16,400 to $133,700 in gross monthly iOS in-app revenue. While free to download and ad-free on Google Play, access beyond initial cases requires a "Pro" subscription listed on iOS at $12.99 monthly, $19.99 or $59.99 for three months, or $89.99 annually.

Gameplay Loop, Content Structure, and Scoring Mechanics

Empirio's core positioning—*"Most financial education tells you what to think. We let your decisions teach you how you think"*—replaces textbook theory with experiential roleplay. Users progress through a standardized **6-Step Framework** inside every business case:

1. **Enter a Real Business Situation:** The user steps into a chronological crossroads faced by a historical or contemporary decision-maker (e.g., at Coca-Cola, Netflix, Volkswagen, Airbnb, Estée Lauder, Novo Nordisk, Nintendo Switch 2, or the Spanish Empire) and reviews the context, constraints, and stakes.
2. **Learn the Concept:** A short visual/animated video lesson ("mini-movie" with a documentary feel, upgraded in v2.2.0 with a new video player supporting hold-for-2x playback speed) explains the strategic or financial framework behind the decision.
3. **Explore the Data:** Users inspect interactive evidence and "fact files" to weigh trade-offs independently.
4. **Make Your Call:** At interactive checkpoints, the narrative pauses and presents real choices with real consequences and no trick questions.
5. **See What Actually Happened:** The app reveals historical numbers, financial outcomes, second-order effects, and lasting lessons, alongside anonymized peer benchmarks (e.g., showing *"70% of users chose this action"*).
6. **Discover Your Patterns:** The user's choice updates their "Business Intuition Score" and plots their decision profile across **10 Foundational Dimensions of Financial Thinking**: *Time Horizon*, *Success Metric*, *Risk Appetite*, *Concentration*, *Leadership*, *Decision Style*, *Social Proof*, *Commitment*, *Value Signal*, and *Root Cause*.

Content is organized across **6 Business Domains**: *Investing & Capital*, *Business Models & Strategy*, *Marketing & Brand*, *Money & Macro*, *Sales & Revenue*, and *Operations & Scale*. Beyond standalone 5-minute cases, Empirio structures retention around **Curated Case Arcs** (guided deep dives into business themes), customizable daily notifications (morning coffee, lunch break, or late night, introduced in v1.4.0), green completion checkmarks (v2.2.0), and multi-episode **Mini-Series** (introduced in v2.0.0 on September 7, 2026) that unfold in order on the homepage.

Job postings for *AI Native Storyteller* and *Social Media Ads Creator* roles at Smartheads in Bucharest reveal the internal production pipeline: human storytellers identify a historical or business subject, structure the narrative arc (scenes, dialogue, tension), design the interactive layer (where decisions land and what quizzes test), and direct in-house AI tooling for research, drafting, voice synthesis, and asset generation, manually correcting generic AI outputs.

User Complaints and Structural Vulnerabilities in Empirio

A synthesis of user reviews across Google Play, the Apple App Store, and Trustpilot surfaces seven recurring product weaknesses that your free, zero-login architecture can directly overcome:

Vulnerability Category	Evidence from Empirio User Reviews & Technical Audits	Architectural Remediation in Your Application
1. Mandatory Login & OTP Restart Bug	Empirio requires account registration with a name, password, and valid email address. Users report a critical bug where switching to an email app to copy the 30-second verification code causes the mobile app to restart, invalidating the code in an infinite loop.	**Zero-Login Local-First State:** Eliminate authentication entirely. The app boots directly into gameplay; all user progress persists locally in MMKV and SQLite / IndexedDB.
2. AI Video Bandwidth, Glitches & Hallucinations	Reviewers complain that AI-generated videos fail to play on weak cellular connections, bottleneck delivery bandwidth, cannot be skipped or paused, and contain distracting visual errors (e.g., an Estée Lauder factory worker placing a box on top of a product on an assembly line).	**Interactive Data Cards & Vector Charts:** Replace heavy AI video streams with instant-loading written narratives, SVG financial charts, and interactive "fact files" that work offline.
3. Unverified AI Claims & Lack of Sources	Users express concern that because material is AI-generated and lacks listed sources, readers cannot verify whether stories are historically factual or contain AI hallucinations.	**Immutable Source Provenance:** Every generated scenario embeds clickable footnotes linking directly to the underlying SEC filing (10-K/8-K), press release, or archival article.
4. Spoiler Leaks in Pre-Decision Framing	Multiple reviewers note that the introductory concept lesson gives away the quiz answers before the user is asked, making the structure predictable for experienced business users.	**Anti-Spoiler LLM Validation Gate:** An automated critic pass strips forward-looking hints and outcome leaks from pre-decision context cards prior to JSON publication.
5. Survivorship & Outcome Bias	Critics highlight that cases suffer from outcome bias—celebrating what worked for one company (e.g., Novo Nordisk) while ignoring the nuance of 100 comparable bets	**Counterfactual & Base-Rate Framing:** The synthesis engine pairs historical outcomes with industry base rates and parallel failure cases to teach probabilistic decision-making.

Vulnerability Category	Evidence from Empirio User Reviews & Technical Audits	Architectural Remediation in Your Application
	that failed.	
6. UI Clipping, Lag Spikes & Missing Navigation	Users report clipped text in "fact files" and option cards without scrollbars when font sizes are enlarged (partially patched in v2.2.0), profile page lag spikes, missing search, and no "Start Here" onboarding flow.	**Responsive Scroll Containers & Local Search:** Enforce scrollable flex containers with dynamic type support, SQLite full-text search, and a guided foundational track.
7. Enterprise Skew & Subscription Friction	Small business founders note an over-focus on giant corporations (e.g., Nintendo Switch 2) over small-business needs (customer/talent acquisition, scaling up), alongside auto-renewal billing disputes.	**100% Free Tier + Multi-Scale Taxonomy:** Zero subscriptions or billing friction; automated tagging sorts cases by company stage (Seed/SMB, Growth, Public Enterprise).

2. Zero-Login, Local-First Client Architecture

Eliminating user registration and server-side accounts removes authentication friction, cuts backend user-database hosting costs to zero, and avoids personal data collection (whereas Brainstash/Deepstash collects names, emails, user IDs, and device IDs under EU-US Standard Contractual Clauses and retains deleted account backups for up to 120 days). In a zero-login architecture, the client device serves as the authoritative system of record for the user's decision profile, completed cases, bookmarks, and daily streaks.

Evaluating Local-First Storage Engines

To prevent the profile page lag spikes reported in Empirio, client persistence should be divided into a **Two-Tier Local Storage Engine** matched to data access patterns:

1. **Hot-Path Synchronous Key-Value Store (react-native-mmkv on Mobile / localStorage + In-Memory Cache on Web):** Built in C++ by Tencent and exposed to React Native via the JavaScript Interface (JSI) and Nitro Modules (react-native-mmkv v4, which requires react-native-nitro-modules, React Native 0.76+, and an Expo development build via npx expo prebuild rather than Expo Go), MMKV persists data using OS-level memory-mapped files (mmap). Unlike AsyncStorage, which serializes every call across an asynchronous bridge and waits for a Promise to resolve, MMKV maps the file directly into the app's memory space so getString and set calls execute synchronously in the same JavaScript tick. Because synchronous calls on the JS thread can block animations if misused for large blobs or tight write loops, MMKV should be reserved strictly for small, high-frequency payloads: the user's current 10-dimension psychometric score vector, active streak counters, theme/font-size preferences, and current checkpoint cursor.
2. **Relational & Queryable Store (op-sqlite or expo-sqlite on Mobile / IndexedDB on Web):** Large relational or queryable datasets—such as the catalog of case headers, downloaded scenario JSON payloads, and the user's chronological decision log—belong

in SQLite. In React Native, **op-sqlite** is a JSI binding engineered for maximum query speed, while **expo-sqlite** provides New Architecture support across Android, iOS, and tvOS, a built-in key-value store (expo-sqlite/kv-store), and free SQLCipher encryption options ("sqlcipher": true in op-sqlite or {"useSQLCipher": true} in expo-sqlite). Indexing case titles, company names, domains, and concepts in SQLite allows millisecond queries across thousands of rows and instant local search—directly resolving Empirio users' complaints about the missing search bar and unclear lesson tracking.

Storage Engine	Underlying Mechanism	Primary Use Case in Your App	Performance & Operational Characteristics
react-native-mmkv (v4)	C++ memory-mapped files (mmap) via JSI & Nitro Modules	Active 10-dimension intuition scores, daily streak state, UI/accessibility settings, current slide index	Synchronous same-tick reads/writes; eliminates async state hydration overhead; requires Expo Prebuild / Dev Client
op-sqlite / expo-sqlite	Native SQLite engine bound via JSI	Catalog of case headers, local search index, completed decision history, cached case JSON bundles	Executes indexed relational queries across thousands of records natively; supports optional SQLCipher encryption
IndexedDB	Browser-native asynchronous object store	Web/PWA client persistence for case catalog and user decision history	Asynchronous non-blocking storage supporting offline JSON case bundles on desktop and mobile browsers

On-Device Psychometric Profiling Model

In Empirio, user choices reveal patterns across 10 financial thinking dimensions. To implement this entirely on-device within your internal codebase, each of the 10 dimensions can be modeled as a bipolar continuous spectrum ranging from -100 to +100, initialized at 0 (neutral baseline):

1. **Time Horizon:** Short-Term Liquidity (-100) ↔ Multi-Decade Compounding (+100)
2. **Success Metric:** Unit Profitability & Cash Flow (-100) ↔ Market Share & Top-Line Scale (+100)
3. **Risk Appetite:** Capital Preservation & Hedging (-100) ↔ Asymmetric Moonshot Bets (+100)
4. **Concentration:** Diversified Portfolio (-100) ↔ Single-Core Focus (+100)
5. **Leadership:** Decentralized Autonomy (-100) ↔ Founder-Directed Command (+100)
6. **Decision Style:** Quantitative Data-First (-100) ↔ Contrarian Intuition-First (+100)
7. **Social Proof:** Consensus & Industry Standard (-100) ↔ First-Principles Iconoclasm (+100)
8. **Commitment:** Iterative Optionality & Pivot-Ready (-100) ↔ Irreversible All-In Commitment (+100)
9. *Value Signal:* Cost Efficiency & Utility (-100) ↔ Brand Scarcity & Pricing Power (+100)
10. **Root Cause:** Structural/Macro Attribution (-100) ↔ Execution/Micro Attribution (+100)

In this proposed scoring architecture, whenever a user selects an option at a decision checkpoint, the option's pre-computed weight array (values from -100 to +100 across relevant dimensions) and a scenario complexity multiplier (0.5 to 1.5) update the user's running score in MMKV. Using an exponentially weighted moving average (newScore = (1 - alpha) * oldScore + alpha * (complexity * optionWeight), where alpha = Math.max(0.05, 1 / (decisionsCount + 1))) ensures that a user's first 5–10 cases establish a responsive profile quickly while subsequent decisions refine the profile smoothly without erratic swings.

Anonymous Cross-Device State Recovery (Zero-Account Sync)

Because users never register an email address or password, switching devices would ordinarily reset their progress. You can provide optional, zero-account state portability through a **Client-Side Compact State Token**. In this design, the user's profile—consisting of ten 16-bit integer dimension scores (20 bytes), streak counters (8 bytes), and a compressed bitfield of completed case IDs (e.g., 1,000 cases represented in 125 bytes)—compresses via Deflate/Zstd into roughly 200–250 bytes. Encoded in Base64URL, this compact payload fits inside a standard **QR Code** or copy-pasteable recovery link that a user can scan with a new phone's camera to restore their full history locally without any server-side user table.

3. Live Web Scraping & Historical Crawling Subsystem

To populate the application automatically with both historical business crossroads and breaking corporate strategy moves, the ingestion layer should combine structured regulatory data pipelines with high-precision web content extractors.

Target Source Taxonomy

A common pitfall of generic news scrapers is ingesting opinion commentary rather than concrete operational decisions with measurable financial stakes. Your ingestion taxonomy should prioritize primary regulatory disclosures alongside business journalism and archival repositories across four tiers:

Source Tier	Target Feeds & Repositories	Strategic Business Moves Captured	Proposed Ingestion Mechanism
1. Regulatory & SEC Filings (Primary Truth)	SEC EDGAR (10-K Item 1 Business, Item 1A Risk Factors, Item 7 MD&A; 10-Q; 8-K Current Reports; S-1 IPO filings; Form 4 insider transactions; 13F-HR holdings)	Mergers & acquisitions, executive changes, earnings releases, segment restructurings, audited balance sheet/income/cash flow outcomes	Python edgartools / edgartools-async querying SEC EDGAR filings
2. Corporate Wire & Earnings Releases	Corporate press releases, earnings call releases, and investor relations RSS feeds	Product launches, pricing model pivots, strategic partnerships, supply chain expansions	Asynchronous RSS/Atom feed listeners + rs-trafilatura extraction
3. Business &	Business news portals,	Competitive dynamics,	Scheduled HTTP

Source Tier	Target Feeds & Repositories	Strategic Business Moves Captured	Proposed Ingestion Mechanism
Strategy Journalism	technology trade press, and entrepreneurship/SMB case blogs	startup scaling bottlenecks, small-business customer/talent acquisition case studies	crawlers + rs-trafilatura main content extraction
4. Historical & Archival Repositories	Historical corporate encyclopedias, public business history archives, and classic economic case repositories	Multi-decade corporate turnarounds, historical empire trade networks (e.g., Spanish Empire), classic corporate failures	Batch backfill crawlers with page-type-aware HTML extraction

Regulatory Ingestion via ed[span_125](start_span)[span_125](end_span)gartools

Rather than paying $50+/month for commercial financial APIs or writing custom HTML scrapers for sec.gov, your ingestion workers can integrate **edgartools** (and its async fork edgartools-async), a free, MIT-licensed open-source Python library for SEC EDGAR extraction since 1994 that its documentation benchmarks at 10–30x the speed of alternative libraries. By configuring a compliant SEC User-Agent identity (set_identity("your.name@example.com")), your pipeline can instantiate any public company (Company("NVDA") or Company("MSFT")) and extract structured Python objects without API keys:

1. **Real-Time Corporate Crossroads (Form 8[span_29](start_span)[span_29](end_span)[span_34](start_span)[span_34](end_span)-K):** Access material corporate events as they happen—including acquisitions, earnings releases, and executive changes—via get_filings(form="8-K")[0].obj(), which parses 8-K filings into structured event items with clean full-text extraction.
2. **Deep Strategic Context & Risks (Form 10-K):** Calling tenk = Company(ticker).get_filings(form="10-K")[0].obj() exposes named section accessors—specifically tenk["Item 1"] (tenk.business), tenk["Item 1A"] (tenk.risk_factors), and tenk["Item 7"] (MD&A)—and converts full filings into clean plain text (filing.text()) or structured Markdown (filing.markdown()) optimized for LLM pipelines.
3. **Audited Financial Outcomes (XBRL Statements):** To populate the "Explore the Data" and "See What Actually Happened" steps with verified numbers rather than unverified AI text, tenk.financials and Company(ticker).get_facts() extract XBRL-tagged Balance Sheets, Income Statements, Cash Flows, Revenue (get_revenue()), Net Income (get_net_income()), Total Assets (get_total_assets()), and Free Cash Flow (get_free_ca[span_18](start_span)[span_18](end_span)[span_24](start_span)[span_24](end_span)sh_flow()) directly into pandas DataFrames with standardized cross-company labels.

Web Content Extraction and Boilerplate Removal Benchmarks

When crawling non-SEC web pages (news articles, corporate blogs, documentation, or historical wikis), raw HTML contains navigation menus, advertisements, login prompts, and cookie banners that must be stripped before LLM processing.

An evaluation published via the US Department of Energy's Office of Scientific and Technical Information (OSTI) assessed main content extraction (MCE) libraries using Rouge-Lsum metrics and found that while no single library statistically outperformed all others across every metric, Python's **trafilatura** achieved the highest mean F1 score of **0.937** and mean precision of **0.978** (with a mean recall of **0.920**), while Mozilla's **Readability** recorded a mean F1 of **0.914**, precision of **0.936**, and the highest mean recall of **0.929**. However, the Web Content Extraction Benchmark (**WCXB**—covering 1,497 pages from 1,295 domains across 7 structural page types) demonstrates that while top extractors converge around F1 = 0.91–0.93 on standard news articles, standard Python trafilatura drops to F1 = **0.792** across mixed web page types because product pages encode descriptions in JSON-LD and forum pages wrap posts in CSS classes that article extractors strip as boilerplate. Additionally, standard trafilatura does not render client-side JavaScript, returns flat text rather than typed database rows on catalog pages, and can over-strip captions on very short stub pages (scoring 0.571 vs. Readability's 0.800 on a short 120-character fixture).

To handle diverse business web sources accurately at minimal compute cost, your crawler pipeline should deploy **rs-trafilatura** (available as a Rust crate and Python package):

1. **Page-Type-Aware Heuristic Extraction:** rs-trafilatura uses URL heuristics and HTML signal analysis (JSON-LD @type values, Open Graph tags, and DOM patterns) to apply multi-candidate section merging on corporate service pages, JSON-LD structured data fallbacks on product pages, and framework-specific boilerplate removal on documentation sites. On the 1,497-page WCXB benchmark, rs-trafilatura achieves an F1 score of **0.859** at **44 ms/page** (and **0.893** on a 511-page held-out test set), outperforming both Python trafilatura (**0.792**) and neural extractors MinerU-HTML (**0.827**) and ReaderLM-v2 1.5B (**0.741** at **10,410 ms/page** on an NVIDIA A100 GPU). Related cross-language ports show similar native speed advantages, with rust-trafilatura v2.2.6 processing pages in **10.91 ms/page** versus **24.75 ms/page** for go-trafilatura v2.2.6.
2. **Confidence-Routed Hybrid Fallback:** rs-trafilatura outputs an extraction confidence score per page. The 92% of pages scoring ≥0.80 confidence stay on the fast heuristic path, while the remaining 8% of low-confidence or client-rendered pages can be routed to a headless browser renderer and MinerU-HTML neural extractor, lifting overall held-out F1 to **0.910**.

Deduplication, Compliance, and Dual-Pipeline Orchestration

Because a major corporate event (such as a merger or CEO change) triggers an SEC 8-K filing alongside dozens of news articles, the ingestion subsystem should separate acquisition from extraction and enforce strict compliance and deduplication rules. All HTTP fetchers must respect domain rate limits and robots.txt policies, including mandatory email identification in the SEC EDGAR User-Agent header. Extracted texts are fingerprinted via locality-sensitive hashing (MinHash) and clustered by company entity and event window so that syndicated duplicates are discarded, while complementary sources (e.g., an 8-K filing plus an investigative business article) are merged into a single **Multi-Source Event Dossier**.

Orchestration is divided into two distinct schedules. First, a **Historical Backfill Pipeline** runs in weekly batch jobs to crawl corporate history archives and SEC 10-K filings (1994–present via edgartools), building evergreen historical cases and multi-episode Mini-Series. Second, a **Live Incremental Pipeline** runs every 6 hours to poll breaking 8-K filings and business news RSS feeds, automatically generating "Latest Move" scenarios as corporate events unfold.

4. Internal Classification & Scenario-Synthesis Engine

The core intelligence of your application resides in the internal codebase that transforms a raw **Multi-Source Event Dossier** into a balanced, spoiler-free, playable business case.

Stage 1: Entity Recognition and Chronological Fact-Graph Extraction

To prevent the outcome bias and unverified claims noted by Empirio reviewers, the synthesis engine never asks an LLM to write a story in a single unstructured pass. Instead, an extraction pass parses the source dossier into a **Chronological Fact Graph** split strictly around the historical **Decision Timestamp (T0)**:

1. **Pre-Decision Context (t ≤ T0):** Extracts the protagonist company, executive role, industry sector, company scale (*Small Business/Startup*, *Mid-Market*, or *Enterprise*), operational constraints known strictly at or before T0, and 3–5 verifiable numerical metrics from the source text or XBRL tables to populate Step 3 ("Explore the Data").
2. **Decision Crossroads (t = T0):** Reconstructs three plausible strategic choices: (Option A) the actual historical path chosen by leadership, (Option B) the primary alternative debated by the board or analysts at T0, and (Option C) a viable contrarian or conservative strategic option.
3. **Post-Decision Outcome (t > T0):** Extracts the verified financial outcome (revenue, net income, or market share changes from subsequent 10-K/10-Q filings or archival records), second-order consequences, and industry base-rate context to populate Step 5 ("See What Actually Happened").

Stage 2: Algorithmic Sorting, Tagging, and Indexing Rules

Once the Fact Graph is constructed, a deterministic classification rule engine assigns metadata tags so the client codebase can sort historical versus latest moves automatically:

1. **Temporal Era Sorting (case_era):** Events with a decision date prior to 2000 are tagged HISTORICAL_FOUNDATIONAL (e.g., Spanish Empire, early Coca-Cola); events from 2000 up to 180 days ago with realized multi-quarter outcomes are tagged MODERN_CASE (e.g., Netflix, Airbnb, Volkswagen); and breaking events from the last 180 days are tagged LATEST_MOVE.
2. **Domain & Scale Taxonomy (domain_id & company_scale):** Each case is classified into one of Empirio's 6 domains (*Investing & Capital*, *Business Models & Strategy*, *Marketing & Brand*, *Money & Macro*, *Sales & Revenue*, *Operations & Scale*), plus a dedicated company_scale filter (SMB_STARTUP, MID_MARKET, ENTERPRISE) so small-business founders can filter directly for customer acquisition, hiring, and bootstrapping cases.
3. **Difficulty & Onboarding Assignment (difficulty_tier):** Calculated from financial metric density and strategic complexity into TIER_1_FOUNDATIONAL (which automatically feeds the structured "Start Here" onboarding curriculum requested by users), TIER_2_OPERATOR, and TIER_3_EXECUTIVE (tailored to experienced leaders who found Empirio's pacing too basic).

Stage 3: Multi-Agent LLM Synthesis and Anti-Spoiler Quality Gates

The synthesis pipeline enforces structured outputs across three sequential stages:

1. **Scenario Architect Pass:** Drafts Step 1 (Situation Brief), Step 2 (Concept Lesson), Step 3 (Interactive Fact Files), and Step 4 (Checkpoints) using only the Pre-Decision Context ($t \leq T0$). Crucially, Step 2 explains the strategic framework using a neutral parallel example so it never gives away the protagonist's decision.
2. **Psychometric Actuary Pass:** Assigns the 10-dimension weight vector (-100 to +100) to each option and drafts Step 5 (Historical Outcome & Base-Rate Counterfactual) and Step 6 (Pattern Discovery) using the Post-Decision Outcome ($t > T0$).
3. **Adversarial Critic & Anti-Spoiler Gate:** An automated verification script and critic pass check four mandatory conditions before publishing: (a) **No Spoiler Leaks** in Steps 1–3 (preventing the answer-telegraphing flaw reported on Trustpilot and the App Store), (b) **Option Plausibility Balance** (ensuring distractor choices are realistic executive trade-offs), (c) **Numerical Source Grounding** (verifying that every financial figure maps to a cited SEC filing or source article URL), and (d) **Base-Rate Nuance** (ensuring Step 5 contextualizes whether a high-risk bet succeeded against industry odds, addressing outcome-bias complaints).

Standardized Playable Scenario JSON Schema Specification

Every validated case is serialized into a self-contained JSON document following the schema specification below, enabling the client app to render rich interactive cases offline without video streaming bottlenecks:

JSON Path / Object Key	Data Type	Pedagogical & Architectural Purpose
case_id, schema_version, published_at	String, ISO-8601	Unique identifier and delta-sync version timestamp for local SQLite catalog reconciliation.
classification.{case_era, domain_id, company_scale, difficulty_tier, series_id}	Object (Enums)	Drives automatic client sorting across Historical vs. Latest Moves, the 6 Business Domains, SMB vs. Enterprise scale, the "Start Here" onboarding track, and Mini-Series episode ordering.
sources[].{s[span_127](start_span)[span_127](end_span)ource_id, type, title, url, verified_date}	Array<Object>	Immutable citations linking directly to SEC EDGAR 10-K/8-K filings or source articles so users can verify every claim.
steps.step_1_situation.{headline, context_markdown, stakes_summary}	Object	**Step 1 (Enter a Real Business Situation):** Presents the executive role, historical date, operational constraints, and financial stakes.
steps.step_2_concept.{framework_name, visual_diagram_type, explanation_markdown, executive_quote}	Object	**Step 2 (Learn the Concept):** Renders a crisp SVG framework diagram, written lesson, and motivating leader quote (requested by reviewers) without leaking the checkpoint answer.
steps.step_3_fact_files[].{tab_title, chart_type, metrics[], deep_dive_text}	Array<Object>	**Step 3 (Explore the Data):** Interactive evidence tabs, native SVG financial charts, and scrollable deep-dive text drawers for users wanting richer written detail.
steps.step_4_checkpoints[].{prompt, complexity_multiplier, options[].{label, trad[span_97](start_span)[span_97](end_span)[span_102](start_span)[span_102](end_span)eo ff_preview, dimension_weights}}	Array<Object>	**Step 4 (Make Your Call):** Decision checkpoints where each option carries explicit trade-offs and a 10-key weight dictionary (TIME_HORIZON through ROOT_CAUSE).
steps.step_5_outcome.{historical_choice_id, actual_outcome_markdown, second_order_effects[], base_rate_counterfactual}	Object	**Step 5 (See What Actually Happened):** Reveals the real decision, XBRL-verified numbers, second-order effects, and industry base rates to counter survivorship bias.

5. Complete Technical Stack, Zero-Cost Edge Delivery & Deployment Roadmap

To satisfy your requirement that the application remain **free with zero login** while scaling to unlimited unauthenticated users at near-zero hosting cost, the runtime system should use a **Static-Edge Content Delivery Model** backed by **Cloudflare R2**.

Decoupled Backend and Cloudflare R2 Capacity Analysis

Instead of running a live backend API server that queries a database on every user session, your synthesis pipeline runs only when ingesting new business moves, uploading static .json case bundles and a lightweight catalog_index.json manifest to **Cloudflare R2**.
In 2026, Cloudflare R2's permanent free tier includes:

1. **10 GB-month** of Standard storage.
2. **1,000,000 Class A operations** per month (writes and lists).
3. **10,000,000 Class B operations** per month (reads).
4. **$0.00 egress bandwidth fees** across all storage classes.

Under this architecture, a typical multi-step structured JSON case (containing markdown narrative, chart arrays, 10-dimension weights, and source citations) requires roughly 15 KB uncompressed (~4 KB compressed). Storing an expansive library of **10,000 historical and live business cases** consumes approximately 150 MB (10,000 × 15 KB)—just **1.5% of R2's 10 GB free storage allowance**. Publishing 20 newly synthesized cases per day plus daily manifest updates requires fewer than 700 Class A write operations per month (<0.07% of the 1 million Class A free tier), while 10 million free Class B reads fronted by Cloudflare CDN caching allow the application to serve hundreds of thousands of unauthenticated users at **$0/month in storage and bandwidth costs**. If you wish to include Empirio's aggregated peer benchmark feature (e.g., *"70% of users chose this action"*), a lightweight serverless endpoint can accept anonymous {case_id, option_id} tallies without user identifiers and periodically write the aggregated percentages into a static peer_stats.json file on R2.

End-to-End Technical Stack Specification

Architectural Layer	Recommended Technology	Role & Justification	Licensing & Cost Profile
Client Framework (Mobile & Web)	**React Native (0.76+ with Expo Prebuild)** + **React Native Web**	Single TypeScript codebase compiling to native iOS, native Android, and web browsers	Free / Open Source
Client Hot-Path Storage	**react-native-mmkv (v4)** via JSI & Nitro Modules	Synchronous same-tick persistence of the 10-dimension intuition score vector, UI preferences, and active streak	Free (MIT License)
Client Relational &	**op-sqlite / expo-sqlite**	Offline-first storage of	Free (MIT License)

Architectural Layer	Recommended Technology	Role & Justification	Licensing & Cost Profile
Search DB		downloaded case packs, instant local text search across cases, and chronological user decision log	
UI & Interactive Charts	**Responsive Scroll Containers** + **SVG Charting**	Prevents Empirio's text-clipping bugs under enlarged system font settings and replaces heavy AI videos with interactive charts	Free / Open Source
Regulatory Filing Scraper	**edgartools / edgartools-async** (Python 3.10–3.13)	Direct extraction of SEC 10-K (Business, Risk Factors, MD&A), 10-Q, 8-K corporate events, and XBRL financial statements	Free (MIT License, no API key or subscription required)
Web Scraper & Boilerplate Cleaner	**Async HTTP / Headless Browser** + **rs-t[span_38](start_span)[span_38](end_span)[span_40](start_span)[span_40](end_span)[span_42](start_span)[span_42](end_span)rafilatura**	High-speed (44 ms/page) page-type-aware extraction of news wires and archives with 0.859–0.910 F1 accuracy	Free (Apache 2.0 / Open Source)
Scenario Synthesis & QA Pipeline	**Python + Structured Schema LLM Pipeline**	Runs offline in scheduled batch jobs to extract Fact Graphs, synthesize 6-step JSON scenarios, and execute Anti-Spoiler Critic checks	Pay-per-token only during batch ingestion ($0 runtime per user)
Content Delivery & Object Storage	**Cloudflare R2**	Hosts catalog_index.json and compressed scenario bundles with zero egress fees	$0 within Free Tier (10 GB storage, 1M Class A, 10M Class B ops/mo)

Phased Implementation Roadmap

1. **Phase 1: Core Schema & Automated Ingestion Pipeline (Weeks 1–2):** Define the JSON validation schemas for the ChronologicalFactGraph and the 6-step playable scenario format. Implement the Python ingestion workers integrating edgartools (for 10-K and 8-K filings) and rs-trafilatura (for business news RSS feeds and historical corporate

archives). Build the three-stage synthesis pipeline with the Anti-Spoiler Critic Gate and generate an initial seed library of 100 historical cases (covering both enterprise titans and small-business scaling challenges) plus 20 live 8-K dilemmas.

2. **Phase 2: Zero-Egress Edge Storage & Delta Manifests (Week 3):** Provision a Cloudflare R2 bucket and configure public read access with immutable cache-control headers. Build the manifest compiler that outputs catalog_index.json (containing lightweight case metadata for sorting by era, domain, company scale, difficulty tier, and Mini-Series arc) alongside compressed weekly case bundles, automated via a 6-hour scheduled cron workflow.
3. **Phase 3: Zero-Login Client & Local-First Storage Engine (Weeks 4–6):** Initialize the React Native (0.76+ Expo Prebuild) client with react-nati[span_31](start_span)[span_31](end_span)[span_36](start_span)[span_36](end_span)ve-mmkv v4 and op-sqlite. Build the delta-sync client module that pulls new case bundles from Cloudflare R2 into local SQLite storage for offline play and full-text search. Implement the 6-Step Case Player UI using scrollable containers tested against enlarged accessibility fonts (preventing Empirio's text-clipping defect), interactive SVG data charts, clickable source footnotes, the on-device 10-dimension scoring engine in MMKV, and the QR Code state export/restore utility.
4. **Phase 4: Curated Case Arcs, "Start Here" Track & Peer Telemetry (Weeks 7–8):** Configure automated thematic clustering to group related cases into multi-episode **Mini-Series**, designate 10 foundational cases for the guided **"Start Here" Onboarding Path**, and deploy the optional anonymous choice-tally endpoint so Step 5 displays peer decision percentages alongside verified historical outcomes.

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